OSCILLATIONS, WAVES AND OPTICS

PHYSICS FOR CHEMISTRY STUDENTS

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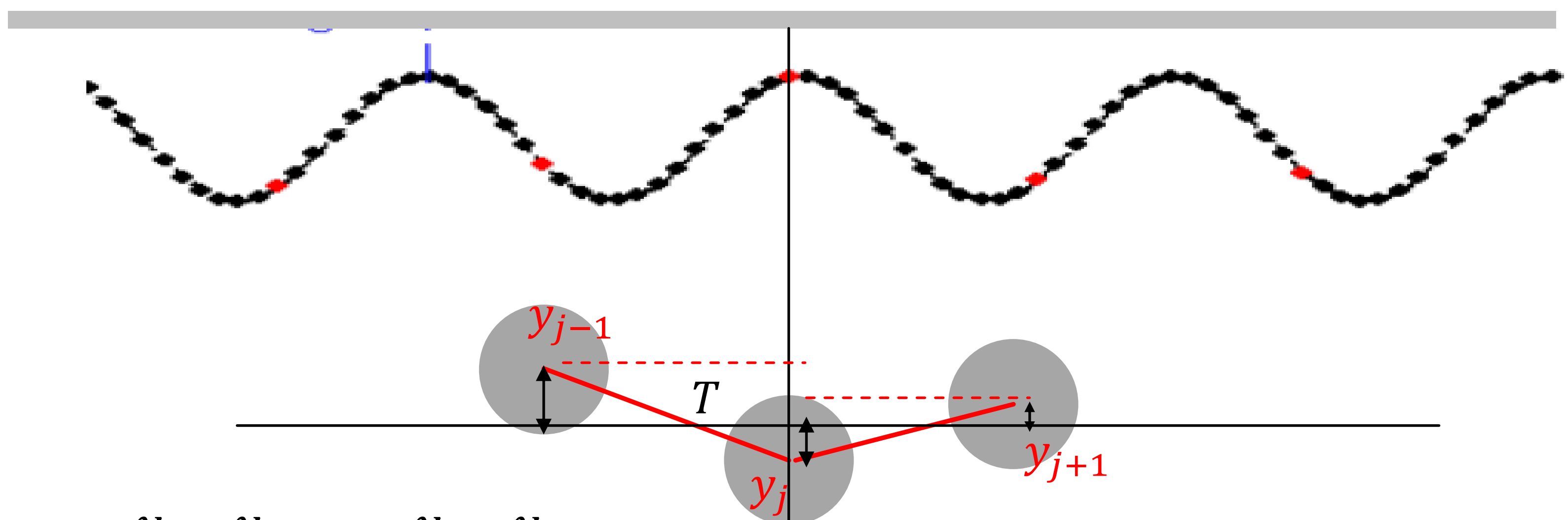
REVIEW

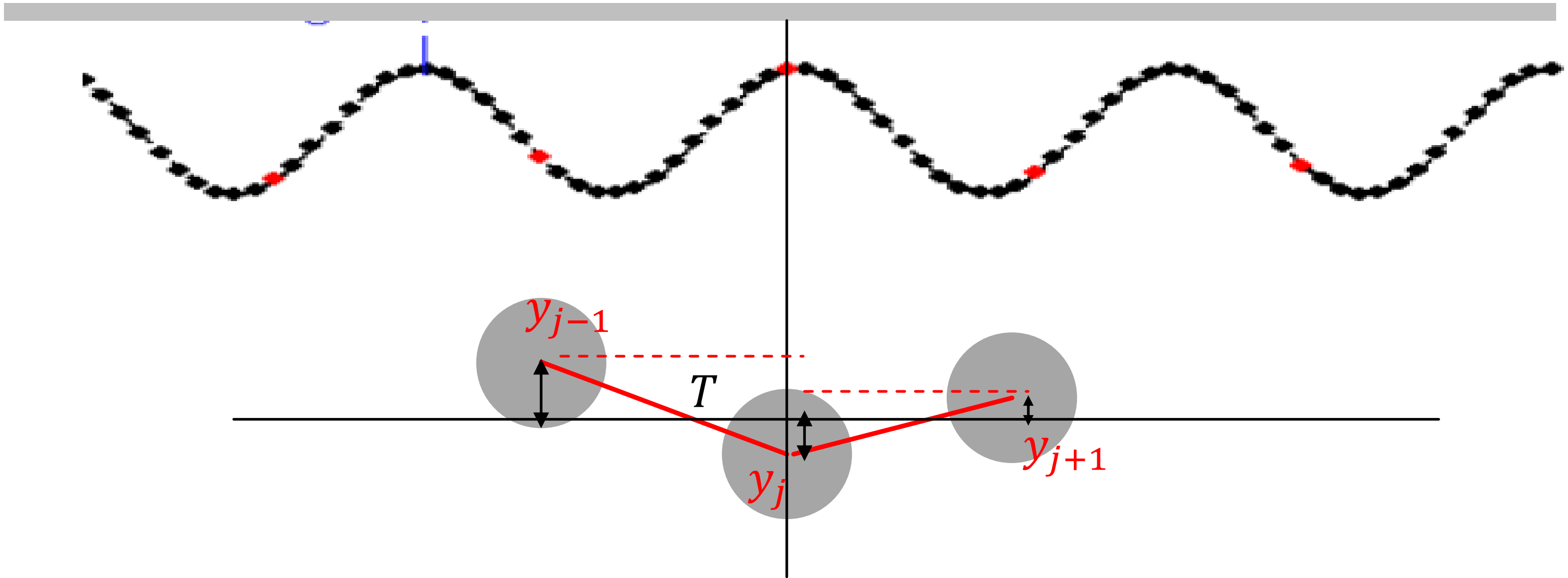


$$m\ddot{y}_j = -T\frac{y_j - y_{j-1}}{a} - T\frac{y_j - y_{j+1}}{a}$$

$$\ddot{\boldsymbol{Y}} = -\boldsymbol{W}\boldsymbol{Y} \qquad \boldsymbol{W} = \begin{pmatrix} \ddots & & 0 & & 0 \\ & 2\omega_0^2 & -\omega_0^2 & 0 & \\ 0 & -\omega_0^2 & 2\omega_0^2 & -\omega_0^2 & 0 \\ & 0 & -\omega_0^2 & 2\omega_0^2 & \\ 0 & & 0 & & \ddots \end{pmatrix}$$

Normal modes: $\boldsymbol{Y} = \boldsymbol{A}e^{i\omega_m t}$ $\quad m = 1, 2,3 \ldots$

REVIEW



$$\ddot{y}_j = \left[-T \frac{\delta y}{\delta x}\bigg|_{x=x_j} + T \frac{\delta y}{\delta x}\bigg|_{x=x_{j+1}} \right] / (\rho \cdot \delta x) = \frac{T}{\rho}\left(y'\big|_{x+\delta x} - y'\big|_{x} \right) / \delta x = \frac{T}{\rho} y''\Big|_{x}$$

$$v_p^2 = \frac{T}{\rho} \rightarrow \boxed{\frac{\partial^2 y}{\partial t^2} = v_p^2 \frac{\partial^2 y}{\partial x^2}}$$ **wave equation**

REVIEW

1D Free Waves: $$\frac{\partial^2 u}{\partial t^2} = v_p^2 \frac{\partial^2 u}{\partial x^2} (-\infty < x < \infty, t > 0)$$

initial condition: $u(0,x) = \varphi(x); \quad \dot{u}(0,x) = \phi(x)$

general solution: $u(t,x) = f\left(x + v_p t\right) + g(x - v_p t)$

$$u(0,x) = f(x) + g(x) = \varphi(x)$$

$$\dot{u}(0,x) = v_p f' - v_p g' = \phi(x) \qquad f(x) - g(x) = \frac{1}{v_p}\int_0^x \phi(\xi) d(\xi) + c$$

$$u(t,x) = f\left(x + v_p t\right) + g(x - v_p t) = \frac{\varphi\left(x + v_p t\right) + \varphi(x - v_p t)}{2} + \frac{1}{2v_p}\int_{x - v_p t}^{x + v_p t} \phi(\xi)\, d\xi$$

d'Alembert's formula

REVIEW

$$\frac{\partial^2 u}{\partial t^2} = v_p^2 \frac{\partial^2 u}{\partial x^2}$$

Dispersion relation: $v_p = \frac{\omega_m}{k_m}$

*m*th normal mode for continuous wave equation

$$u_m(x,t) = c_m \sin(k_m x + \varphi)\sin(\omega_m t + \phi)$$

Separation of variables

PDE→ODE

General solution

$$u(x,t) = \sum_m c_m \sin(k_m x + \varphi)\sin(\omega_m t + \phi)$$

Normal Modes !

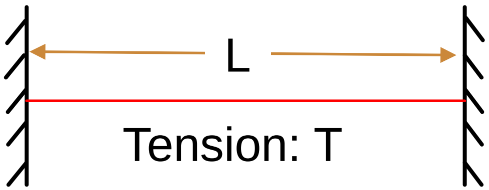
c_m k_m ω_m are unknown

REVIEW

*m*th normal mode for continuous wave equation

$$u_m(x,t) = c_m \sin(k_m x + \varphi)\sin(\omega_m t + \phi)$$

In specific cases: Two fix ends



$$u_m(0,t) = c_m \sin(\varphi)\sin(\omega_m t + \phi) = 0$$

$$\varphi = n\pi$$

$$u_m(L,t) = c_m \sin(k_m L + \varphi)\sin(\omega_m t + \phi) = 0$$

Boundary conditions:

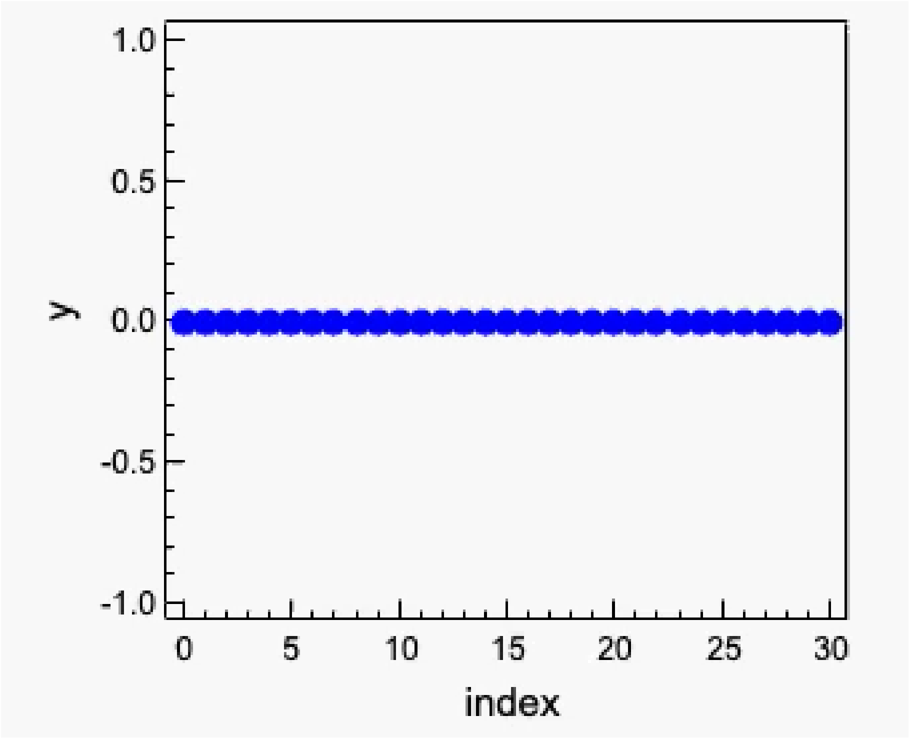
$$u(0,t) = 0$$

$$u(L,t) = 0$$

$$k_m L + \varphi = n'\pi$$

$$k_m = m\pi/L$$

REVIEW



$$k_m = \frac{m\pi}{L} \qquad \lambda_m = \frac{2\pi}{k_m} = \frac{2L}{m}$$

$$\lambda_1 = 2L \qquad k_1 = \frac{\pi}{L}$$

$$\lambda_2 = L \qquad k_2 = \frac{2\pi}{L}$$

$$\lambda_3 = \frac{2}{3}L \qquad k_3 = \frac{3\pi}{L}$$

$$\lambda_4 = \frac{1}{2}L \qquad k_3 = \frac{4\pi}{L}$$

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$$\lambda_5 = \frac{2}{5}L \qquad k_5 = \frac{5\pi}{L}$$

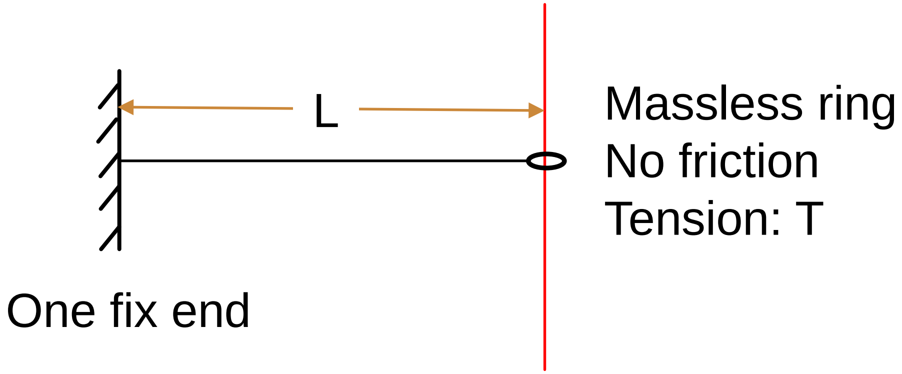
3.3 BOUNDARY CONDITIONS

*m*th normal mode for continuous wave equation

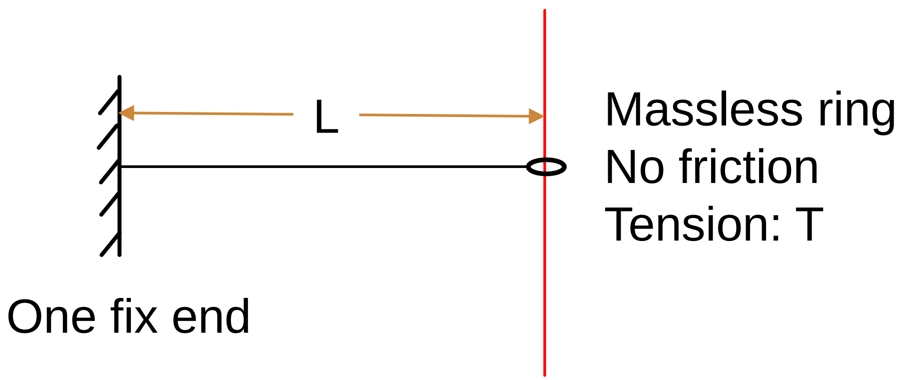
$$u_m(x,t) = c_m \sin(k_m x + \varphi)\sin(\omega_m t + \phi)$$

General solution of the wave equation

$$u(x,t) = \sum_m c_m \sin(k_m x + \varphi)\sin(\omega_m t + \phi)$$



3.3 BOUNDARY CONDITIONS



Boundary conditions:

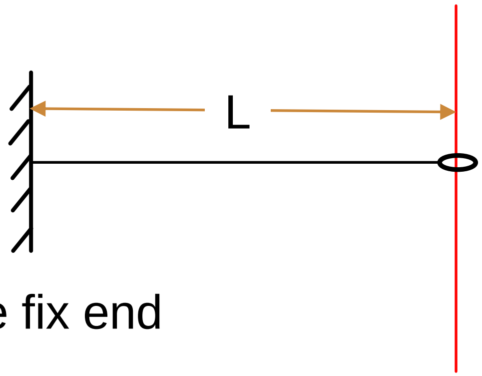
$$u(0,t)=0$$

$$u'(L,t)=0$$

$$u(0,t)=0 \qquad u_m(0,t)=c_m\sin(\varphi)\sin(\omega_m t+\phi)=0$$

$$\varphi=n\pi$$

3.3 BOUNDARY CONDITIONS



One fix end

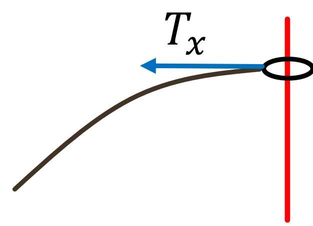
Massless ring
No friction
Tension: T

Boundary conditions:

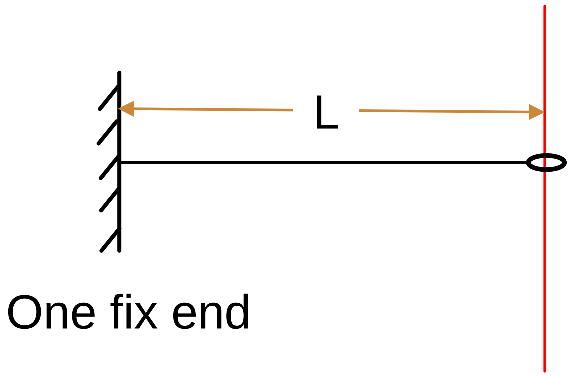
$$u(0,t) = 0$$

$$u'(L,t) = 0$$

$$u'(L,t) = 0$$



3.3 BOUNDARY CONDITIONS



Massless ring
No friction
Tension: T

Boundary conditions:

$$u(0,t) = 0$$

$$u'(L,t) = 0$$

$$u(0,t) = 0 \qquad \varphi = n\pi$$

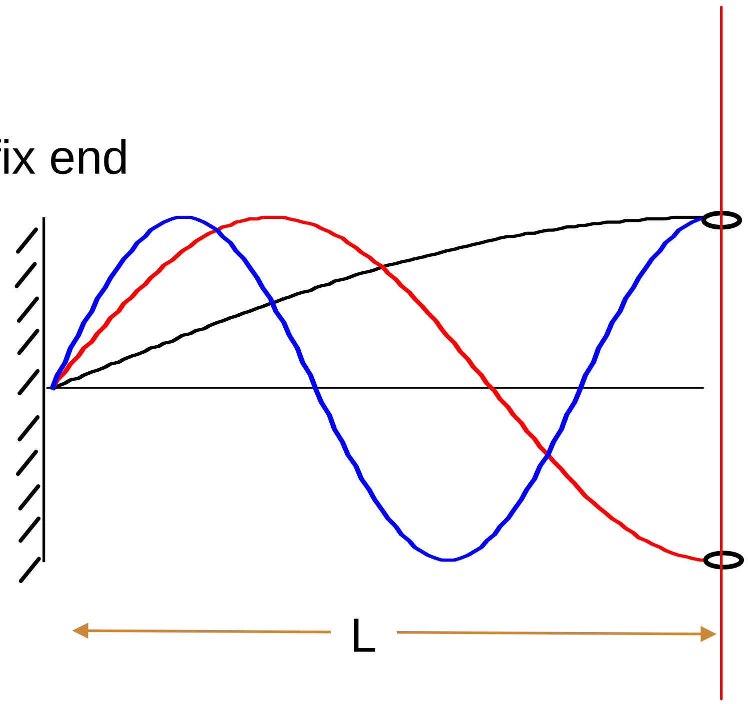
$$u'(L,t) = 0 \qquad u'_m(L,t) = k_m c_m \cos(k_m L) \sin(\omega_m t + \phi) = 0$$

$$k_m L = (2m+1)\frac{\pi}{2} \qquad k_m = (2m+1)\frac{\pi}{2L}$$

General solution: $u_m(x,t) = c_m \sin(k_m x) \sin(\omega_m t + \phi)$

3.3 BOUNDARY CONDITIONS

One fix end



$$k_m = (2m+1)\frac{\pi}{2L}$$

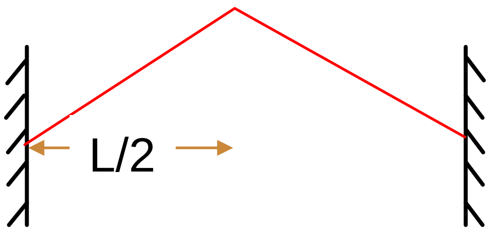
$$\lambda_m = \frac{2\pi}{k_m} = \frac{4L}{2m+1}$$

$$\lambda_0 = 4L \quad k_0 = \frac{\pi}{2L}$$

$$\lambda_1 = \frac{4}{3}L \quad k_1 = \frac{3\pi}{2L}$$

$$\lambda_2 = \frac{4}{5}L \quad k_2 = \frac{5\pi}{2L}$$

3.4 FOURIER SERIES



Initial condition t=0:

General solution

$$u(x,t) = \sum_m c_m \sin(k_m x + \varphi)\sin(\omega_m t + \phi)$$

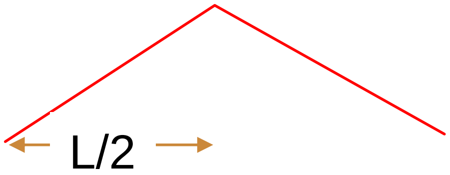
Boundary conditions: $k_m = \dfrac{m\pi}{L}$

Initial condition

$$u(x,0) = \sum_m c_m \sin(k_m x + \varphi)\sin(\phi)$$

Can $\sum_m c_m \sin(k_m x + \varphi)\sin(\phi)$ construct $u(x,0)$?

3.4 FOURIER SERIES



General solution

$$u(x,t) = \sum_m c_m \sin(k_m x + \varphi)\sin(\omega_m t + \phi)$$

Initial condition t=0: $u(x,0), \dot{u}(x,0) = 0$

Initial condition

$$u(x,0) = \sum_m c_m \sin(k_m x + \varphi)\sin(\phi)$$

Can

$$\sum_m c_m \sin(k_m x + \varphi)\sin(\phi)$$

construct $u(x,0)$, and how?

3.4 FOURIER SERIES

$f(x)$ is a periodic function; the period is 2π; $f(x)$ is differentiable.

Fourier series:

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(nx) + b_n \sin(nx)\,,\, n = 1,2,3\,...$$

Orthogonality of trigonometric polynomials : $m \neq n$

$$\int_{-\pi}^{\pi} \cos(nx)\cos(mx)dx = \int_{-\pi}^{\pi} \sin(nx)\sin(mx)dx = 0$$

$m = n$

$$\int_{-\pi}^{\pi} \cos(nx)\sin(mx)dx = 0$$

$$\int_{-\pi}^{\pi} \sin^2(nx)\,dx = \int_{-\pi}^{\pi} \cos^2(mx)\,dx = \pi$$

3.4 FOURIER SERIES

$f(x)$ is a periodic function; the period is 2π; $f(x)$ is differentiable.

Fourier series:

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(nx) + b_n \sin(nx)\,, n = 1,2,3 \ldots$$

$$\int_{-\pi}^{\pi} f(x) \cos(mx)\, dx =$$

$$\int_{-\pi}^{\pi} \frac{a_0}{2} \cos(mx)\, dx + \sum_{n=1}^{\infty} \int_{-\pi}^{\pi} [a_n \cos(nx) \cos(mx) + b_n \sin(nx) \cos(mx)] dx$$

$$= a_m \pi$$

$$a_m \pi = \int_{-\pi}^{\pi} f(x) \cos(mx)\, dx$$

3.4 FOURIER SERIES

$f(x)$ is a periodic function; the period is 2π; $f(x)$ is differentiable.

Fourier series:

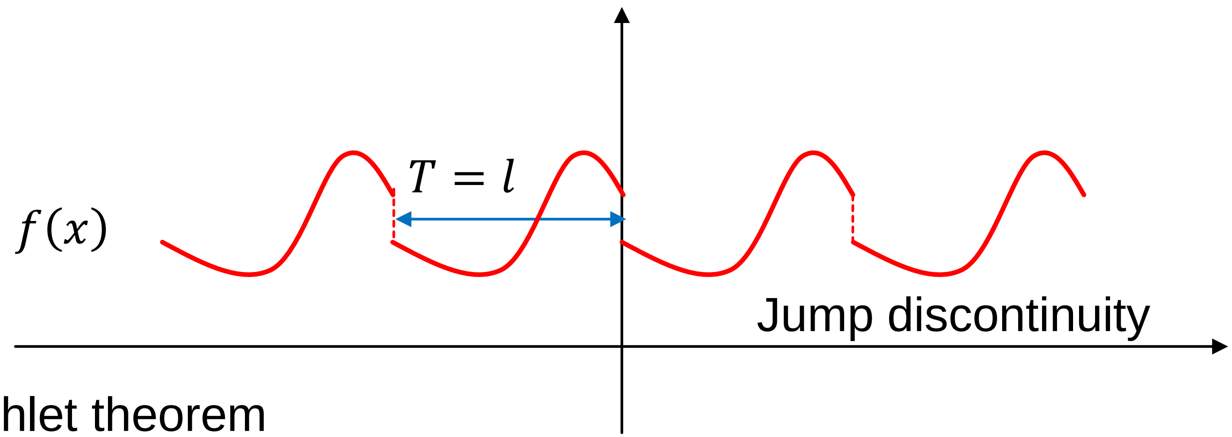
$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(nx) + b_n \sin(nx) \text{ , } n = 1,2,3 \ldots$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos(nx)\, dx \text{ , } n = 1,2,3 \ldots$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin(nx) dx \text{ , } n = 1,2,3 \ldots$$

3.4 FOURIER SERIES

$f(x)$ is a periodic function; the period is l



Dirichlet theorem

3.4 FOURIER SERIES

$f(x)$ is a periodic function, and the period is l

$$t = \frac{2\pi}{l}x \qquad f(x) = f\left(\frac{l}{2\pi}t\right) = g(t)$$

$$g(t+2\pi) = f\left[\frac{l}{2\pi}(t+2\pi)\right] = f\left[\frac{l}{2\pi}t + l\right] = f\left(\frac{l}{2\pi}t\right) = g(t)$$

Fourier series:

$$f(x) = g(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(nt) + b_n \sin(nt)\,, n = 1,2,3\ldots$$

$$a_n = \frac{1}{\pi}\int_{t=0}^{t=2\pi} g(t)\cos(nt)\,dt = \frac{1}{\pi}\int_{x=0}^{x=l} f(x)\cos\left(\frac{2\pi}{l}nx\right)\frac{2\pi}{l}dx = \frac{2}{l}\int_0^l f(x)\cos\left(\frac{2\pi}{l}nx\right)dx$$

3.4 FOURIER SERIES

$f(x)$ is periodic function, and the period is l

$$f(x) \approx \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{2\pi}{l} nx\right) + b_n \sin\left(\frac{2\pi}{l} nx\right)$$

Fourier series:

$$a_n = \frac{2}{l}\int_0^l f(x)\cos\left(\frac{2\pi}{l} nx\right) dx$$

$$b_n = \frac{2}{l}\int_0^l f(x)\sin\left(\frac{2\pi}{l} nx\right) dx$$

3.4 FOURIER SERIES

$f(x)$ is odd function, and the period is l; [$f(x)\sin\left(\frac{2\pi}{l}nx\right)$ is even]

$$f(x) \approx \sum_{n=1}^{\infty} b_n \sin\left(\frac{2\pi}{l}nx\right) \qquad a_n = 0$$

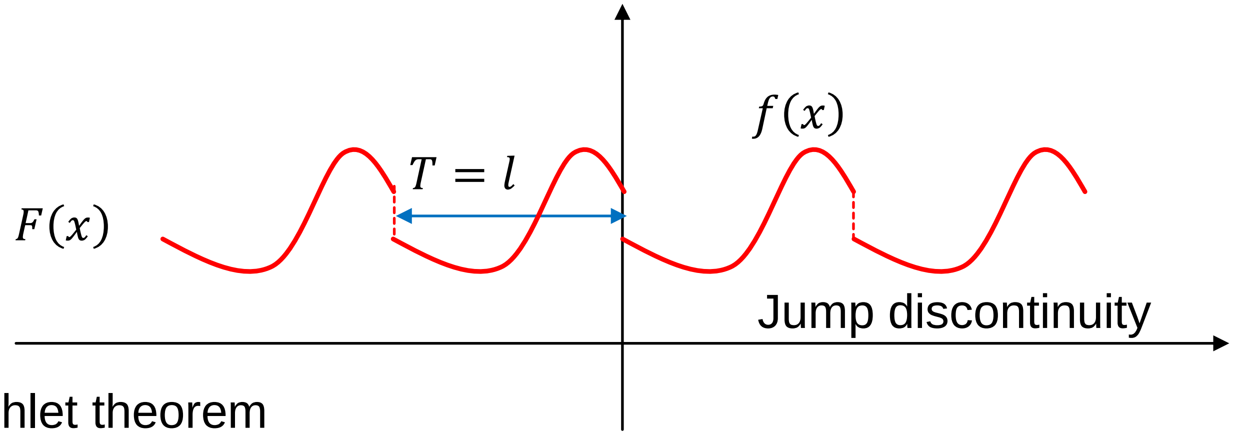
$$b_n = \frac{4}{l}\int_0^{l/2} f(x)\sin\left(\frac{2\pi}{l}nx\right)dx$$

$f(x)$ is even function, and the period is l; [$f(x)\cos\left(\frac{2\pi}{l}nx\right)$ is even]

$$f(x) \approx \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{2\pi}{l}nx\right) \qquad b_n = 0$$

$$a_n = \frac{4}{l}\int_0^{l/2} f(x)\cos\left(\frac{2\pi}{l}nx\right)dx$$

3.4 FOURIER SERIES

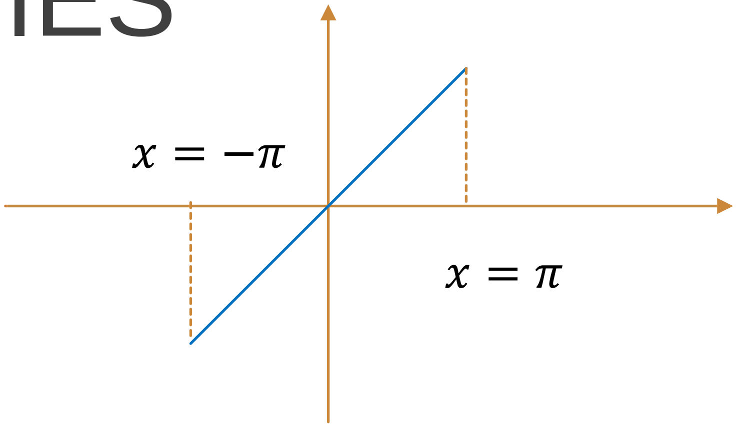


Dirichlet theorem

$$F(x) \approx \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{2\pi}{l} nx\right) + b_n \sin\left(\frac{2\pi}{l} nx\right), n = 1,2,3 \ldots$$

$$f(x) \approx \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{2\pi}{l} nx\right) + b_n \sin\left(\frac{2\pi}{l} nx\right), x \in (0, l)$$

3.4 FOURIER SERIES



$$s(x) = \frac{x}{\pi}, \quad \text{for } -\pi < x < \pi,$$

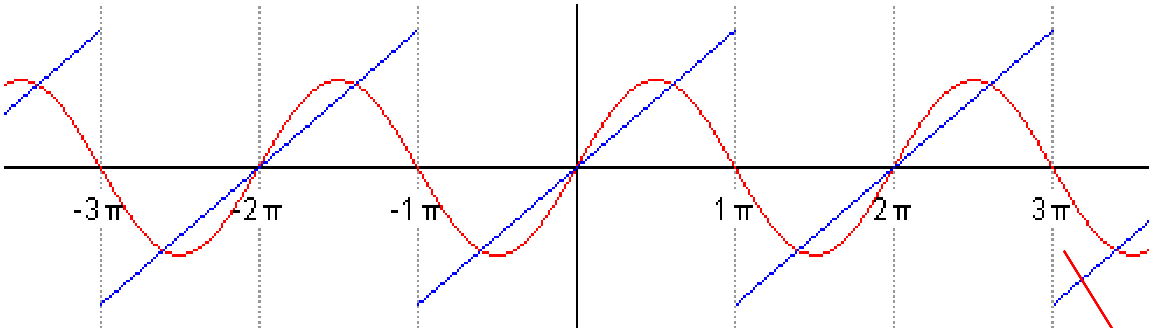
$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} s(x) \cos(nx)\, dx = 0, \quad n \geq 0.$$

$$\begin{aligned} b_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} s(x) \sin(nx)\, dx \\ &= -\frac{2}{\pi n} \cos(n\pi) + \frac{2}{\pi^2 n^2} \sin(n\pi) \\ &= \frac{2\,(-1)^{n+1}}{\pi n}, \quad n \geq 1. \end{aligned}$$

Integration by parts

$$\int u\, dv = uv - \int v\, du.$$

3.4 FOURIER SERIES



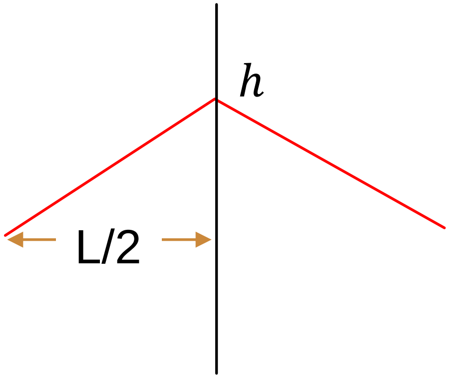
$$s(x) = \frac{x}{\pi}, \quad \text{for} -\pi < x < \pi,$$

the Fourier series converges to 0

$$s(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left[a_n \cos(nx) + b_n \sin(nx) \right]$$

$$= \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} \sin(nx), \quad \text{for} \quad x - \pi \notin 2\pi\mathbb{Z}.$$

3.5 INITIAL CONDITIONS



Dirichlet theorem

General solution

$$u(x,t) = \sum_m c_m \sin(k_m x + \varphi)\sin(\omega_m t + \phi)$$

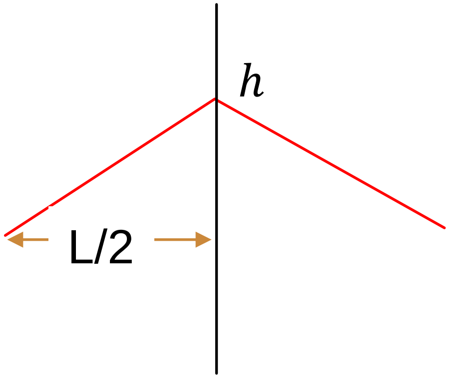
initial condition $t = 0$

$$\dot{u}(x,0) = \sum_m c_m \omega_m \sin(k_m x + \varphi)\cos(\phi) = 0$$

$$\phi = \frac{n\pi}{2}, n = 1,3,5 \ldots$$

$$u(x,t) = \sum_m c_m \sin(k_m x + \varphi)\sin(\omega_m t + \frac{n\pi}{2})$$

3.5 INITIAL CONDITIONS



General solution

$$u(x,t) = \sum_m c_m \sin(k_m x + \varphi)\sin(\omega_m t + \frac{n\pi}{2})$$

$n = 1,3,5 \ldots$

initial condition $t = 0$

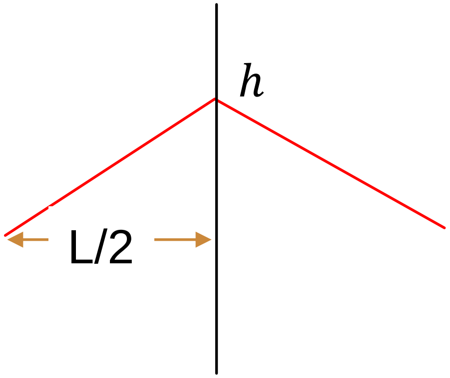
$$u(x,t) = \sum_m c_m \sin(k_m x + \varphi) \cdot 1$$

$k_m = \frac{2\pi}{L}$

$u(x,0)$ is an even function $\quad \varphi = \frac{m'\pi}{2}, m' = 1,3,5 \ldots$

$$u(x,0) = c_0 + \sum_{m=1,\ldots} c_m \cos(\frac{2\pi}{L} m x)$$

3.5 INITIAL CONDITIONS



$$u(x,0) = c_0 + \sum_{m=1,\dots} c_m \cos\left(\frac{2\pi}{L} mx\right)$$

$$u(x,0) = -\frac{2h}{L}x + h; 0 < x < \frac{L}{2}$$

$$u(x,0) = \frac{2h}{L}x + h; -\frac{L}{2} < x < 0$$

$$c_m = \frac{2}{L}\int_0^L u(x,0)\cos\left(\frac{2\pi}{L}mx\right)dx$$

$$= \frac{4}{L}\int_{-L/2}^{0} \frac{2h}{L} x \cos\left(\frac{2\pi}{L}mx\right)dx + \frac{4}{L}\int_{-L/2}^{0} h\cos\left(\frac{2\pi}{L}mx\right)dx$$

3.5 INITIAL CONDITIONS

Integration by parts $\int u dv = uv - \int v du$

$$\int u \frac{dv}{dx} dx = uv - \int v \frac{du}{dx} dx$$

$$\int_{-L/2}^{0} ax \cos\left(\frac{2\pi}{L} mx\right) dx = ax \cdot \frac{L}{m2\pi} \sin\left(\frac{2\pi}{L} mx\right) - \frac{L}{m2\pi} \int_{-L/2}^{0} \sin\left(\frac{2\pi}{L} mx\right) \cdot a dx$$

$$= \left[ax \cdot \frac{L}{m2\pi} \sin\left(\frac{2\pi}{L} mx\right) - a\left(\frac{L}{m2\pi}\right)^2 \cos\left(\frac{2\pi}{L} mx\right)\right]\Bigg|_{-L/2}^{0}$$

$$= 0 - a\left(\frac{L}{m2\pi}\right)^2 \cos(m\pi) + a\left(\frac{L}{m2\pi}\right)^2$$

$$c_m = \frac{4}{L} \int_{-L/2}^{0} \frac{2h}{L} x \cos\left(\frac{2\pi}{L} mx\right) dx = -\frac{8h}{L^2}\left(\frac{L}{m2\pi}\right)^2 \cos(m\pi) + \frac{8h}{L^2}\left(\frac{L}{m2\pi}\right)^2$$

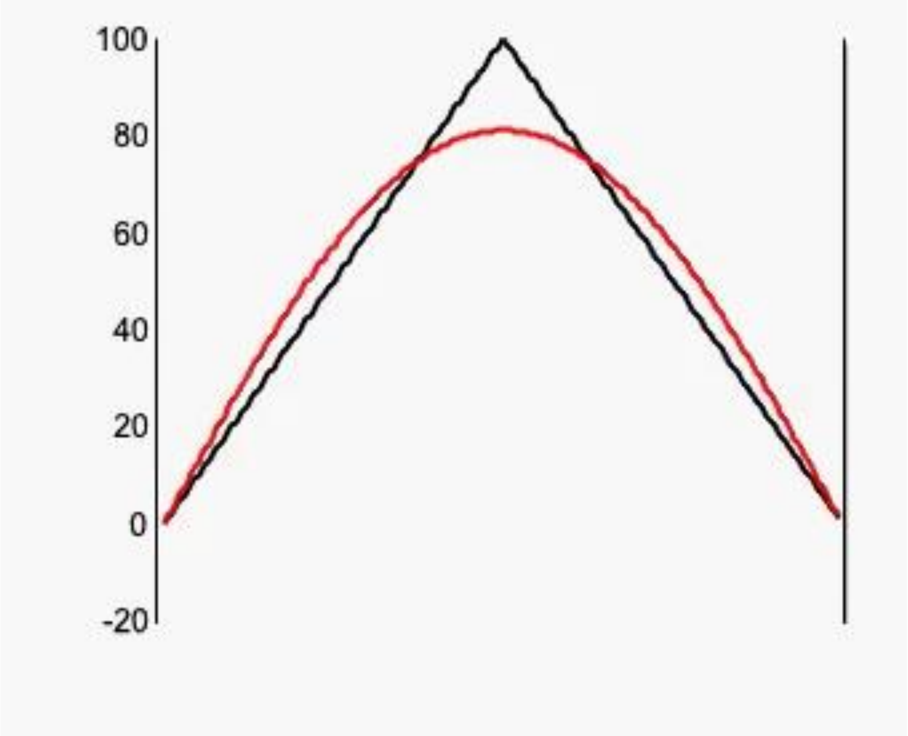
3.5 INITIAL CONDITIONS

$$c_m = -2h\left(\frac{1}{m\pi}\right)^2 \cos(m\pi) + 2h\left(\frac{1}{m\pi}\right)^2$$

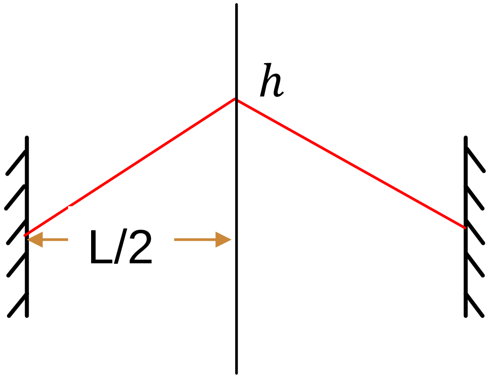
When m is even $\qquad c_m = 0$

When m is odd $\qquad c_m = 4h\left(\frac{1}{m\pi}\right)^2$

$$u(x,0) = \sum_{m=1,3,5\ldots} 4h\left(\frac{1}{m\pi}\right)^2 \cos(\frac{2\pi}{L}mx)$$



3.5 INITIAL CONDITIONS



General solution

$$u(x,t) = \sum_m c_m \sin(k_m x + \varphi)\sin(\omega_m t + \phi)$$

Boundary conditions:

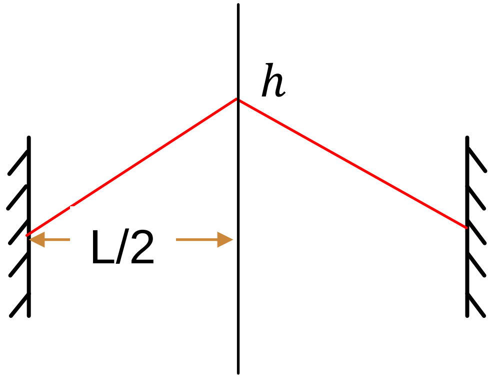
$$u_m\left(-\frac{L}{2}, t\right) = c_m \sin\left(-k_m \frac{L}{2} + \varphi\right)\sin(\omega_m t + \phi) = 0$$

$$u_m\left(\frac{L}{2}, t\right) = c_m \sin\left(k_m \frac{L}{2} + \varphi\right)\sin(\omega_m t + \phi) = 0$$

$$-k_m \frac{L}{2} + \varphi = n\pi \qquad k_m = \frac{m\pi}{L}$$

$$k_m \frac{L}{2} + \varphi = l\pi \qquad \varphi = \frac{\pi}{2}$$

3.5 INITIAL CONDITIONS



General solution

$$u(x,t) = \sum_m c_m \sin(k_m x + \varphi)\sin(\omega_m t + \phi)$$

Boundary conditions:

$$u(x,\mathrm{t}) = \sum_{m=0,1,2\ldots} c_m \sin(\frac{m\pi}{L}x + \varphi)\sin(\omega_m t + \phi)$$

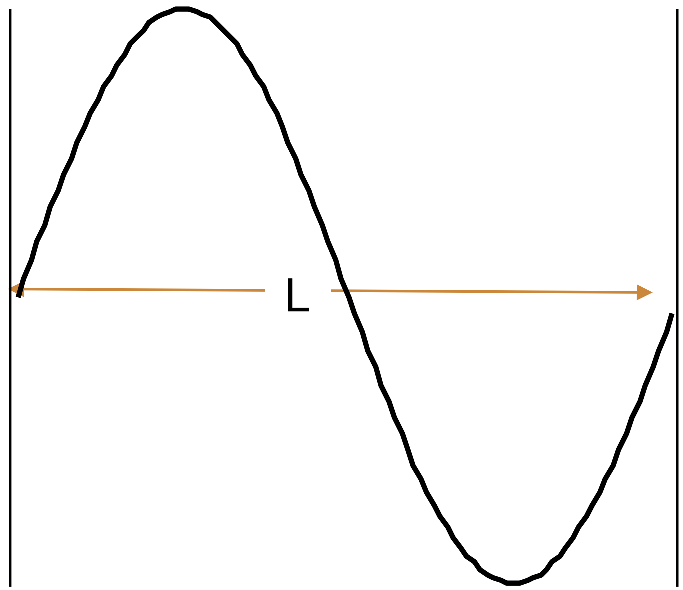
$$\varphi = \frac{m'\pi}{2}$$

Initial conditions:

When m is even $\quad c_m = 0$

$$u(x,t) = \sum_{m=1,3,5\ldots} 4h\left(\frac{1}{m\pi}\right)^2 \cos\left(\frac{2\pi}{L}mx\right)\cos(\omega_m t)$$

3.5 INITIAL CONDITIONS



Initial waveform is a normal mode (m)

$$c_m = \frac{2}{L}\int_0^L u(x,0)\sin\left(\frac{2\pi}{L}mx\right)dx$$

$$\int_0^L \sin(k_m x)\sin(k_n x)dx = 0, m \neq n$$

Orthogonality of trigonometric polynomials

Orthogonality of normal modes

MINI PROJECTS

可演示的物理实验设计:

a.实验目的（振动与波）

b.原理说明

c.实验装置

d.介绍如何演示

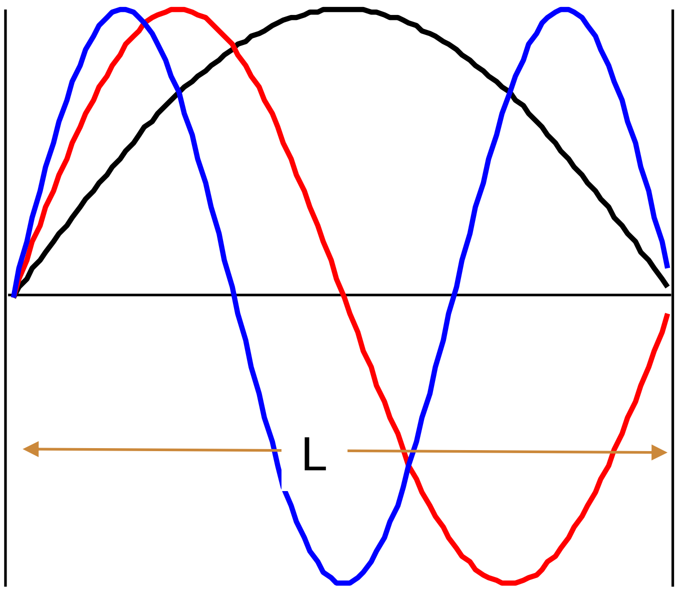
e.预期实验结果+分析

分组+PPT讲解（10月26日）+小论文

MINI PROJECTS

*m*th normal mode for continuous wave equation

$$u_m(x,t) = c_m \sin(k_m x + \varphi)\sin(\omega_m t + \phi)$$



$$\omega = k\sqrt{\frac{T}{\rho}}$$

$$k_m = \frac{m\pi}{L} \qquad \lambda_m = \frac{2\pi}{k_m} = \frac{2L}{m}$$

$$\lambda_1 = 2L \quad k_1 = \frac{\pi}{L} \quad \omega_1$$

$$\lambda_2 = L \quad k_2 = \frac{2\pi}{L} \quad \omega_2$$

$$\lambda_3 = \frac{2}{3}L \quad k_3 = \frac{3\pi}{L} \quad \omega_3$$

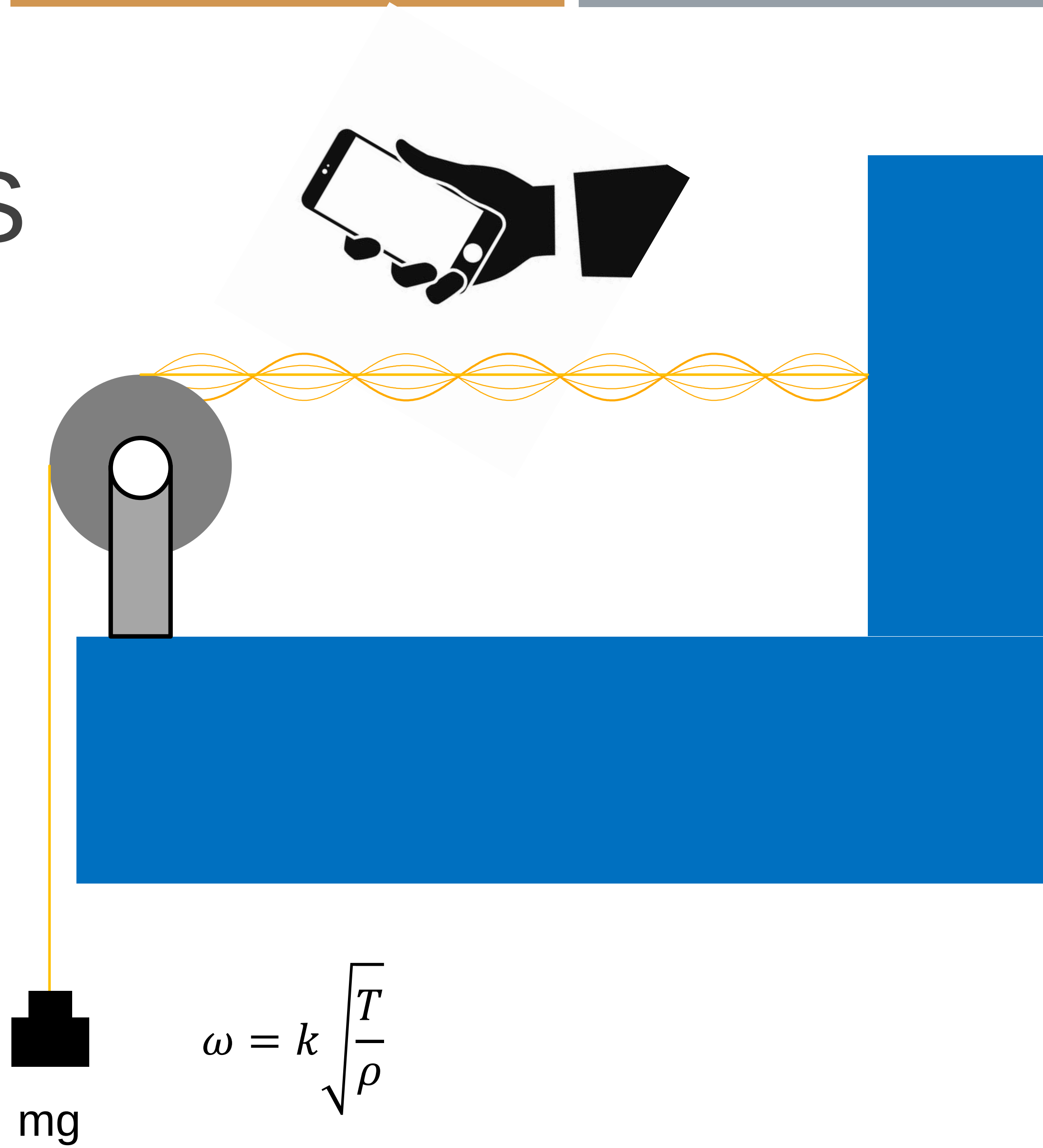
$$\lambda_4 = \frac{1}{2}L \quad k_3 = \frac{4\pi}{L} \quad \omega_4$$

⋮

EXAMPLES

1. 定滑轮
2. 钢丝（铜线、皮筋）
3. 砝码
4. 录音设备
5. 傅里叶变化（软件）

1. 声音频率取值分立
2. 基频与m的关系
3. …



$$\omega = k\sqrt{\frac{T}{\rho}}$$